Learning Augmentation Weights Through One Classifier Update in Corruption-Robust Continual Learning

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Abstract

How does the share of the original image in an augmented view affect corruption robustness in continual learning? We compare chain-aware mixing and a recurrent MetaMix baseline at five original-image weights on Split-CIFAR-10. Across three seeds, classifier InfoNCE improves mean corrupted class-incremental accuracy over Jensen–Shannon (JS) consistency at all five weights for chain-aware mixing and at four for MetaMix. At the strongest tested augmentation ($m = 0.1$), the two InfoNCE classifiers reach 43.89% and 41.66%, respectively. At $m = 0.3$ their order reverses (42.28% and 42.51%). The AugMix reference leads on corrupted task-incremental accuracy at 84.15%. The results link classifier consistency to mixture strength and show why conclusions about mixer design depend on which metric and strength are compared.

1. Introduction

An augmentation controller can reduce disagreement between a clean image and its transformed views by assigning more weight to the original. At the extreme, all views coincide. This presents a difficulty when prediction consistency is used to train the controller: a smaller loss can reflect an easier view even when the classifier has learned little about corruptions. In continual learning, the choice of views also affects the update shared by current examples and examples replayed from earlier tasks.

AugMix draws and blends short stochastic augmentation chains [3]. MetaMix adapts mixture weights using a recurrent summary of the learner and splits training operations into pseudo-seen and pseudo-unseen pools [4]. We study a modification with two linked choices. First, the mixer observes how each sampled seen chain changes the image's representation and prediction. Second, independent random weights determine the pseudo-unseen views on which the classifier update is scored. This design lets the mixer influence the outer loss only through its choice of training views.

We compare the two controllers at matched original-image weights $m \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. At each weight we cross controller choice with classifier consistency. Table 1 reports all four evaluation metrics so the reader can compare controllers within a row and follow each controller as augmentation strength changes.

2. One-Step Mixing

Following MetaMix [4], we concatenate the current batch (x_t, y_t) and a labeled replay batch (x_r, y_r) into (x_b, y_b). At step t, the pseudo-seen pool S_t produces two mixed views x'_{b1}, x'_{b2} of this batch; the disjoint pseudo-unseen pool U_t produces $\hat{x}_{b1}, \hat{x}_{b2}$. For either pool, a mixed view of one image x has the form

$$\tilde{x} = mx + (1 - m) \sum_{k=1}^{3} w_k a_k(x), \qquad \sum_{k=1}^{3} w_k = 1, \quad (1)$$

where m is the original-image weight, a_k is a sampled chain, and $w_k \geq 0$ is its weight. We write θ for classifier parameters and ϕ for mixer parameters.

What the controller observes. The MetaMix baseline uses an LSTM conditioned on current and replay features, replay loss and gradient alignment. For a given example it shares the three chain weights across views and pools. One variant also learns m and the other uses the prescribed sweep value. Resampling chains at fixed learner context leaves its weights unchanged.

The chain-aware controller uses the prescribed m for both operation pools. For each seen chain it concatenates the change in pooled features with the change in class probabilities relative to the clean input. A shared scorer combines this detached descriptor with the recurrent state, then a softmax over the three scores supplies weights separately for each example and seen view. For each pseudo-unseen view we draw $w \sim \mathrm{Dirichlet}(1, 1, 1)$ independently of the controller and keep the draw fixed throughout the batch's mixer updates.

What the classifier learns. Write $\ell_\theta(x, y) = \mathrm{CE}(f_\theta(x), y)$ for mean batch cross-entropy, where f_θ

Loss	m	Clean class-IL		Corrupt class-IL		Clean task-IL		Corrupt task-IL	
		CA	MM	CA	MM	CA	MM	CA	MM
JS	0.1	43.99	**45.69**	38.20	**39.23**	86.05	**86.56**	80.48	**80.88**
	0.3	**42.61**	40.78	**36.33**	34.31	**87.56**	86.61	**81.29**	79.82
	0.5	**44.59**	43.51	**36.56**	36.38	86.26	**86.89**	79.14	**80.05**
	0.7	**42.94**	41.69	**34.96**	34.26	**86.65**	86.12	**79.63**	79.07
	0.9	42.87	**45.60**	33.41	**36.34**	85.58	**86.55**	77.64	**78.88**
InfoNCE	0.1	**51.87**	49.25	**43.89**	41.66	**88.46**	87.74	**83.24**	82.17
	0.3	**50.27**	50.14	42.28	**42.51**	**87.87**	87.43	**81.94**	81.85
	0.5	**46.26**	45.71	**37.49**	37.27	**87.12**	**87.12**	80.92	**81.01**
	0.7	**46.08**	45.63	**36.72**	36.20	86.96	**87.07**	**79.64**	79.59
	0.9	43.33	**44.01**	34.18	**34.60**	**86.55**	86.53	**79.43**	79.19

Table 1. Mean final accuracy (%) over seeds 17, 23, and 43. Column pairs compare chain-aware (CA) and MetaMix (MM) at the same original-image weight and classifier consistency loss. For each loss and m, bold marks the higher CA or MM mean within each metric pair. Ties at the displayed precision are both bold.

returns logits. A separate replay draw (x_d, z_d) supplies stored logits for distillation. Our DER++ loss [1] is

$$L_{\mathrm{DER++}}(\theta) = \ell_\theta(x_t, y_t) + 0.5\,\ell_\theta(x_r, y_r) + 0.2\,\mathrm{MSE}(f_\theta(x_d), z_d). \tag{2}$$

MSE averages over examples and logit coordinates. Replay terms are zero when memory is empty. The classifier objective is

$$L_S(\theta, \phi) = L_{\mathrm{DER++}}(\theta) + C_\theta(x_b, x'_{b1}, x'_{b2}). \tag{3}$$

Here C_θ is either JS or class-aware InfoNCE; the latter also uses the labels y_b. The dependence on ϕ enters through the mixed views.

For three aligned batches v_0, v_1, v_2 of size n, let p_{ij} denote the classifier's probability vector for example i in view v_j, and let $\bar{p}_i = (p_{i0} + p_{i1} + p_{i2})/3$. We use

$$\mathcal{L}_{\mathrm{JS},\theta}(v_0, v_1, v_2) = \frac{1}{3n} \sum_{i=1}^{n} \sum_{j=0}^{2} \mathrm{KL}(p_{ij} \,\|\, \bar{p}_i). \tag{4}$$

For InfoNCE, index the $3n$ views by a and let u_a be the normalized pooled feature. The positive set P_a contains the other two views of the same batch entry; N_a contains all views with a different class label. At temperature $\tau = 0.1$,

$$\mathcal{L}_{\mathrm{NCE},\theta} = -\frac{1}{3n} \sum_a \frac{1}{2} \sum_{b \in P_a} \log \frac{\exp(u_a^\top u_b/\tau)}{\sum_{c \in P_a \cup N_a} \exp(u_a^\top u_c/\tau)}. \tag{5}$$

How the outer loss reaches the mixer. After one virtual classifier step of size α, we evaluate clean classification and pseudo-unseen consistency on the same batch:

$$\theta^+(\phi) = \theta - \alpha \nabla_\theta L_S(\theta, \phi), \tag{6}$$

$$L_U(\theta^+, \phi) = \ell_{\theta^+}(x_b, y_b) + \mathcal{L}_{\mathrm{JS},\theta^+}(x_b, \hat{x}_{b1}, \hat{x}_{b2}). \tag{7}$$

The mixer minimizes $J(\phi) = L_U(\theta^+(\phi), \phi)$. Holding the sampled batch, incoming state and outer BatchNorm buffers fixed, the chain rule gives

$$\nabla_\phi J = (D_\phi \theta^+)^\top \nabla_{\theta^+} L_U + \partial_\phi L_U. \tag{8}$$

The partial derivative holds the virtual classifier fixed. For chain-aware mixing the outer images are independent of ϕ, so this term vanishes. For MetaMix the pseudo-unseen images depend on learned chain weights even when m is fixed, so a direct JS gradient can remain. Clean cross-entropy reaches either mixer through the classifier update. At $m = 1$ the three inputs coincide and their JS is zero under deterministic evaluation. This identity describes the consistency term alone.

Three mixer updates recompute the virtual classifier from the same starting parameters. The classifier commits the step computed with the incoming mixer once per batch. This is the full derivative of one SGD step. The MetaMix baseline uses the project's interpretation of the published solver.

3. Experimental Protocol

The 288,746-parameter SmallCNN learns five successive CIFAR-10 class pairs, $(0, 1)$ through $(8, 9)$, with ten epochs per task and a 500-example reservoir. Each batch has up to 32 current examples and up to 32 labeled replay examples. The classifier and mixer use constant SGD learning rates of 0.003 and 0.005. Each run processes 500,000 current-image presentations in 15,650 classifier updates.

Final evaluation uses 10,000 clean test images and 15 CIFAR-10-C corruptions at five severities [2]. Each corruption and severity receives equal weight. Class-IL predicts among all ten outputs. Task-IL restricts prediction to the true class pair. The pseudo-unseen operations used in training are sampled training transformations. CIFAR-10-C supplies only the test measurements.

References and aggregation. DER++ with AugMix samples two views from the full nine-operation pool, with independent $m \sim \mathrm{Beta}(1, 1)$ and chain weights drawn from $\mathrm{Dirichlet}(1, 1, 1)$. We evaluate its original JS coefficient of 12 and a coefficient of one matched to the JS classifiers. Both use the same replay and classifier update budgets. Each mean gives equal weight to seeds 17, 23, and 43. We report sample SD across seeds in Table 5 and compute differences from the displayed means.

4. Results

Mixture strength across four metrics. For chain-aware InfoNCE all four means decline as m increases (Table 1). The largest class-IL difference occurs on corruptions: 43.89% at $m = 0.1$ versus 34.18% at $m = 0.9$. MetaMix InfoNCE reaches its highest class-IL means at $m = 0.3$ and its highest task-IL means at $m = 0.1$. With JS, the highest corrupted class-IL mean occurs at $m = 0.1$ for both mixers, although intermediate values give a less regular ordering.

What the chain-aware design adds. With classifier InfoNCE and $m = 0.1$, chain-aware leads on all four metrics. At $m = 0.3$ MetaMix leads on corrupted class-IL by 0.23 points, while chain-aware leads on the other three metrics. At $m = 0.5$ their clean task-IL means coincide to two decimals. The comparison changes both seen-chain scoring and pseudo-unseen mixture construction.

Interaction with classifier consistency. Table 2 holds the mixer and mixture strength fixed when comparing classifier losses. Chain-aware gains 5.69 and 5.95 points on corrupted class-IL from InfoNCE at $m = 0.1$ and 0.3, compared with 0.93 at $m = 0.5$. MetaMix gains 8.20 points at $m = 0.3$ but loses 1.74 at $m = 0.9$. The gains also depend on task information: at $m = 0.7$ the chain-aware task-IL gains are only 0.31 points clean and 0.01 corrupted.

		Class-IL		Task-IL	
Mixer	m	Clean	Corrupt	Clean	Corrupt
Chain-aware	0.1	+7.88	+5.69	+2.41	+2.76
	0.3	+7.66	+5.95	+0.31	+0.65
	0.5	+1.67	+0.93	+0.86	+1.78
	0.7	+3.14	+1.76	+0.31	+0.01
	0.9	+0.46	+0.77	+0.97	+1.79
MetaMix	0.1	+3.56	+2.43	+1.18	+1.29
	0.3	+9.36	+8.20	+0.82	+2.03
	0.5	+2.20	+0.89	+0.23	+0.96
	0.7	+3.94	+1.94	+0.95	+0.52
	0.9	−1.59	−1.74	−0.02	+0.31

Table 2. Change in mean accuracy (percentage points) when the classifier switches from JS to InfoNCE, holding the mixer and m fixed.

Learned and random mixture strength. Tables 1 and 3 show how the fixed settings compare with learned m and random AugMix. Within MetaMix with classifier InfoNCE, corrupted class-IL is 35.26% with learned m, 42.51% at fixed $m = 0.3$, and 34.60% at fixed $m = 0.9$. Thus the value chosen matters even when m is fixed. AugMix with its original JS coefficient reaches the highest corrupted task-IL mean (84.15%). Reducing that coefficient to one raises clean class-IL from 45.94% to 48.99%, while corrupted class-IL changes from 40.00% to 39.68%. At the matched coefficient AugMix exceeds both controllers' best observed JS means on corrupted class-IL.

		Class-IL		Task-IL	
Mixer	Loss	Clean	Corrupt	Clean	Corrupt
MetaMix, learned m	JS	46.16	36.43	86.65	79.83
MetaMix, learned m	InfoNCE	45.48	35.26	86.51	79.14
DER++ + AugMix	1× JS	**48.99**	39.68	87.92	81.81
DER++ + AugMix	12× JS	45.94	**40.00**	**88.33**	**84.15**

Table 3. Reference methods with learned or sampled original-image weight. Entries are three-seed mean accuracies (%). Bold marks the highest mean in each metric among these reference methods.

Which corruptions account for the differences? Tables 6 and 7 give the results for each corruption. With InfoNCE and $m = 0.5$, chain-aware mixing exceeds MetaMix on 13 of 15 class-IL corruption means. The largest advantage is on pixelation (39.74% versus 39.10%). MetaMix leads on impulse noise and contrast. Changing the classifier loss within the chain-aware mixer produces larger differences: switching from JS to InfoNCE raises impulse-noise class-IL by 6.63 points but lowers fog class-IL by 3.57 points. When the true task is supplied, the same switch raises accuracy on 14 corruptions, with motion blur the exception (82.33% with JS and 81.72% with InfoNCE).

How concentrated are the learned weights? At $m = 0.1$ the chains supply 90% of each mixed image. For InfoNCE, the most heavily weighted chain receives 38.24% of that share on average with chain-aware mixing and 34.37% with MetaMix (Table 4). The lower entropy for chain-aware mixing likewise indicates a more uneven allocation. These final-epoch measurements establish a difference in weighting behavior at this strength. They do not identify which chains improved learning.

Mixer	Loss	Max. weight	Entropy
Chain-aware	JS	.3603	1.0917
Chain-aware	InfoNCE	.3824	1.0783
MetaMix	JS	.3463	1.0981
MetaMix	InfoNCE	.3437	1.0979

Table 4. Final-epoch weights at $m = 0.1$, averaged across seeds. The largest of the three weights is taken before averaging. Uniform weights give a maximum of $1/3$ and entropy $\ln 3$.

5. Limitations and Conclusion

The most consistent gain on corrupted class-IL is from classifier InfoNCE at stronger augmentation. Chain-aware InfoNCE attains the highest observed mean on this metric, while the advantage over MetaMix changes sign across matched strengths. The maximum is selected from five settings on the reported evaluation. With three seeds, dispersion is substantial: at $m = 0.1$ the two InfoNCE corrupted class-IL means are 43.89 ± 5.68 and 41.66 ± 7.68 (sample SD). Isolating the effect of candidate descriptors would require holding outer-view sampling and per-view weight sharing fixed. The per-corruption breakdown below covers the original six configurations; the concentration measurements cover $m = 0.1$.

Reproducibility. The repository includes the three-seed reporting script, the one-step trainer, and the PBS jobs for the fixed-strength sweep and AugMix coefficient comparison.

References

[1] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: a strong, simple baseline. In *Advances in Neural Information Processing Systems*, 2020.

[2] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019.

[3] Dan Hendrycks, Norman Mu, Ekin D. Cubuk, Barret Zoph, Justin Gilmer, and Balaji Lakshminarayanan. AugMix: A simple data processing method to improve robustness and uncertainty. In *International Conference on Learning Representations*, 2020.

[4] Zhenyi Wang, Li Shen, Donglin Zhan, Qiuling Suo, Yanjun Zhu, Tiehang Duan, and Mingchen Gao. MetaMix: Towards corruption-robust continual learning with temporally self-adaptive data transformation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24521–24531, 2023.

A. Variation across training seeds

Mixer and loss	m	Clean class-IL	Corrupt class-IL	Clean task-IL	Corrupt task-IL
CA JS	0.1	43.99 ± 8.64	38.20 ± 6.12	86.05 ± 2.84	80.48 ± 1.92
	0.3	42.61 ± 5.12	36.33 ± 3.76	87.56 ± 1.01	81.29 ± 1.19
	0.5	44.59 ± 3.42	36.56 ± 3.15	86.26 ± 1.99	79.14 ± 1.77
	0.7	42.94 ± 7.54	34.96 ± 4.90	86.65 ± 1.00	79.63 ± 0.97
	0.9	42.87 ± 9.60	33.41 ± 7.11	85.58 ± 1.18	77.64 ± 1.91
CA InfoNCE	0.1	51.87 ± 7.29	43.89 ± 5.68	88.46 ± 0.72	83.24 ± 0.62
	0.3	50.27 ± 4.64	42.28 ± 3.03	87.87 ± 0.53	81.94 ± 0.29
	0.5	46.26 ± 7.46	37.49 ± 5.95	87.12 ± 1.47	80.92 ± 1.31
	0.7	46.08 ± 4.40	36.72 ± 3.01	86.96 ± 0.68	79.64 ± 1.22
	0.9	43.33 ± 6.16	34.18 ± 4.78	86.55 ± 0.63	79.43 ± 0.93
MM JS	0.1	45.69 ± 7.10	39.23 ± 4.69	86.56 ± 1.79	80.88 ± 1.08
	0.3	40.78 ± 6.02	34.31 ± 5.57	86.61 ± 2.68	79.82 ± 3.24
	0.5	43.51 ± 4.55	36.38 ± 3.30	86.89 ± 0.94	80.05 ± 0.62
	0.7	41.69 ± 4.83	34.26 ± 2.55	86.12 ± 1.35	79.07 ± 0.74
	0.9	45.60 ± 7.86	36.34 ± 4.37	86.55 ± 1.09	78.88 ± 0.86
MM InfoNCE	0.1	49.25 ± 9.09	41.66 ± 7.68	87.74 ± 1.88	82.17 ± 1.92
	0.3	50.14 ± 5.61	42.51 ± 3.66	87.43 ± 1.42	81.85 ± 1.15
	0.5	45.71 ± 8.40	37.27 ± 6.33	87.12 ± 1.80	81.01 ± 1.41
	0.7	45.63 ± 5.83	36.20 ± 3.97	87.07 ± 0.76	79.59 ± 1.31
	0.9	44.01 ± 5.66	34.60 ± 4.26	86.53 ± 0.51	79.19 ± 0.77

Table 5. Fixed-weight sweep, mean accuracy (%) $\pm$ sample SD over three seeds.

B. CIFAR-10-C accuracy by corruption

Each entry averages the five severity levels within a seed and then takes the arithmetic mean across seeds 17, 23 and 43. Overall gives equal weight to each of the 15 corruptions. Class-IL predicts among all ten classes. Task-IL predicts within the true class pair.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	28.76	**33.76**	30.49	33.61	29.49	30.42
Shot Noise	32.53	**36.57**	33.82	36.38	32.92	33.06
Impulse Noise	22.97	29.60	25.64	**29.70**	25.64	27.40
Defocus Blur	**44.85**	43.99	44.19	43.57	42.32	40.62
Glass Blur	34.00	**36.04**	34.17	36.01	35.28	34.27
Motion Blur	**41.58**	38.66	40.59	38.50	38.26	34.97
Zoom Blur	**44.66**	42.77	44.21	42.58	40.41	38.72
Snow	33.57	**36.51**	33.41	36.27	35.99	35.09
Frost	31.30	31.09	30.49	31.01	**32.48**	30.39
Fog	**39.47**	35.90	37.97	35.81	37.59	34.91
Brightness	39.68	40.77	38.47	40.34	**42.22**	40.50
Contrast	**30.16**	28.86	29.45	28.99	29.37	27.61
Elastic Transform	**44.42**	43.45	43.36	42.96	42.78	41.28
Pixelate	37.71	**39.74**	37.20	39.10	37.96	36.84
JPEG Compression	42.79	**44.68**	42.26	44.17	43.75	42.88
Overall	36.56	**37.49**	36.38	37.27	36.43	35.26

Table 6. Class-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	73.07	77.64	75.04	**77.69**	74.89	73.52
Shot Noise	76.24	80.28	77.99	**80.56**	77.69	76.28
Impulse Noise	71.88	76.32	73.34	**76.40**	73.06	72.47
Defocus Blur	84.20	84.62	84.76	**84.79**	83.76	83.58
Glass Blur	73.70	77.05	75.27	**77.34**	75.58	74.83
Motion Blur	82.33	81.72	**83.09**	82.12	81.86	80.39
Zoom Blur	83.15	83.18	**83.75**	83.39	82.30	82.18
Snow	78.97	**81.59**	79.92	81.55	79.90	80.89
Frost	78.34	**80.09**	78.96	**80.09**	79.31	79.12
Fog	80.01	80.95	80.07	**81.00**	80.57	79.14
Brightness	83.97	**84.71**	84.60	84.54	84.56	84.43
Contrast	70.05	70.82	70.28	70.71	**71.21**	68.55
Elastic Transform	83.42	84.15	84.12	**84.36**	83.61	83.65
Pixelate	83.92	**85.66**	84.60	85.55	84.50	83.69
JPEG Compression	83.90	85.01	84.98	**85.10**	84.69	84.36
Overall	79.14	80.92	80.05	**81.01**	79.83	79.14

Table 7. Task-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.